

Streamlined spatial and environmental expression signatures characterize the minimalist duckweed *Wolffia australiana*

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Single-cell genomics permits a new resolution in the examination of molecular and cellular dynamics, allowing global, parallel assessments of cell types and cellular behaviors through development and in response to environmental circumstances, such as interaction with water and the light–dark cycle of the Earth. Here, we leverage the smallest, and possibly most structurally reduced, plant, the semiaquatic *Wolffia australiana*, to understand dynamics of cell expression in these contexts at the whole-plant level. We examined single-cell-resolution RNA-sequencing data and found *Wolffia* cells divide into four principal clusters representing the above- and below-water-situated parenchyma and epidermis. Although these tissues share transcriptomic similarity with model plants, they display distinct adaptations that *Wolffia* has made for the aquatic environment. Within this broad classification, discrete subspecializations are evident, with select cells showing unique transcriptomic signatures associated with developmental maturation and specialized physiologies. Assessing this simplified biological system temporally at two key time-of-day (TOD) transitions, we identify additional TOD-responsive genes previously overlooked in whole-plant transcriptomic approaches and demonstrate that the core circadian clock machinery and its downstream responses can vary in cell-specific manners, even in this simplified system. Distinctions between cell types and their responses to submergence and/or TOD are driven by expression changes of unexpectedly few genes, characterizing *Wolffia* as a highly streamlined organism with the majority of genes dedicated to fundamental cellular processes. *Wolffia* provides a unique opportunity to apply reductionist biology to elucidate signaling functions at the organismal level, for which this work provides a powerful resource.

[Supplemental material is available for this article.]

Multicellular organisms comprise specialized tissues accommodating diverse cell types. This variety is required to achieve the array of functions necessary for an organism to develop and thrive in dynamic environments, and comes about through precise coordination of complex gene regulatory networks that integrate responses to internal and external cues. Plants, in particular, have evolved intricate mechanisms to accommodate diverse biotic and abiotic environmental inputs, leading to changes in their metabolism, physiology, and development at cellular to organismal levels. Understanding commonalities and variations in these responses on a molecular level is vital to begin decoding the fundamental drivers of tissue distinction and how responses to various inputs may be coordinated at the organismal level to achieve a desired phenotype. Advances in single-cell-resolution RNA sequencing (scRNA-seq) have opened the possibilities for systems approaches to query cellular specialization during development and their specific

responses to environmental cues, allowing for transcriptomic profiling at high spatiotemporal resolution (Seyfferth et al. 2021; Denyer and Timmermans 2022). In plants, scRNA-seq studies published to date have largely focused on aspects of *Arabidopsis* development at the organ or tissue level (e.g., Denyer et al. 2019; Kim et al. 2021; Zhang et al. 2021; Shahan et al. 2022). However, although many of these studies have examined in depth, for example, cell lineage progressions within a developing tissue, no studies have yet profiled at the whole-plant level to sufficient depth to enable observations at the system level. This is principally owing to the high level of structural complexity and large cell numbers in most plant models.

A seminal study published in 2017 did profile the transcriptome for every individual cell within the model nematode *Caenorhabditis elegans* (Cao et al. 2017). A key characteristic that enabled this feat is the simplicity of body plan and low number of cells and cell types within this model animal. A plant paralog to such a model can be found in the *Wolffia* genus of the *Lemnaceae* family. These plants, ranging in size from <1 mm to several millimeters, have a

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